Chapter 9

Conclusions

This thesis investigated how financial news can be used to measure the evolution of emerging risks and whether the resulting measures contain information relevant to financial markets. The analysis focused on five risk categories: Cyber Risk, Climate Risk, Pandemic Risk, Supply Chain Risk, and AI-Related Risk. The objective was to move beyond a simple count of news articles and examine different aspects of how risks appear and evolve in the information environment.

The research was developed in two main parts. The first constructed a framework for identifying risk-related articles and measuring changes in their frequency and semantic content. The second applied the resulting indicators to Cyber Risk and investigated their relationship with subsequent weekly equity portfolio excess returns. The empirical results show that the indicators capture different dimensions of risk-related reporting and that, among the measures examined in the financial analysis, Coverage provides the clearest evidence of predictive information.

The starting point was the construction of a consistent news database. The original material covered articles published between 2014 and 2023 by five newspapers: the Financial Times, Wall Street Journal, New York Times, Los Angeles Times, and USA Today. Before using these articles to measure risk-related attention, it was necessary to verify their metadata, standardize their structure, and address duplicated content. This step was important because repeated articles could artificially increase the measured frequency of a risk theme, while inconsistencies in publication dates or sources could affect comparisons over time.

The database initially contained 2,374,643 records in the staging table. After validation, 2,373,999 articles were transferred to the main database. The subsequent identification of 70,964 exact duplicates and 53,780 near-duplicates produced a final analytical corpus of 2,249,255 articles. The resulting dataset provided a common and consistent basis for the classification and measurement procedures developed in the following chapters.

The next task was to identify which articles were related to the five selected risks. This was not treated as a simple assignment of each article to one category. A news article can discuss more than one risk, and its relevance to each theme can vary. The classification framework therefore assigned continuous thematic relevance scores to article–risk pairs.

Three approaches were developed: keyword-based scoring, weakly-supervised scoring, and supervised scoring. The keyword approach provided a transparent lexical baseline, but its results depended on the presence of predefined expressions. Weak supervision introduced contextual labeling rules to capture more specific evidence of thematic relevance. The supervised approach used TF–IDF representations and logistic regression to learn associations between textual features and the risk categories.

The comparison of these approaches was important because the identification of relevant articles directly affects the subsequent indicators. An overly restrictive classification can exclude articles that discuss a risk without using its most common terminology. Conversely, an overly broad classification can include unrelated reporting and distort the measured evolution of the theme. The use of manually reviewed observations and classification diagnostics therefore served to evaluate the reliability of the thematic scores before their application to the complete corpus.

The final scoring procedure evaluated each of the 2,249,255 articles against all five risk categories. This produced 11,246,275 article–theme combinations for each methodology and 33,738,825 method–article–theme observations across the three approaches. These scores made it possible to measure thematic relevance systematically over a large collection of financial news.

Thematic classification alone, however, could not describe how the information contained in the articles changed over time. Two articles may concern the same risk while discussing substantially different events or developments. Similarly, articles may use different words to describe closely related information. For this reason, the thesis introduced document embeddings as a second representation of the corpus.

The embeddings mapped articles into a common semantic space, allowing their textual content to be compared through vector similarity. This representation complemented thematic relevance scoring. While classification identified the risk categories associated with each article, embeddings provided information about the similarity between articles and about changes in the content associated with a theme.

Combining these two representations made it possible to construct four indicators: Novelty, Semantic Drift, Coverage, and Burst Detection. Each was designed to capture a different characteristic of risk emergence.

Novelty measured how different incoming articles were from previously available information. Semantic Drift measured changes in the semantic representation of a risk category over time. Coverage measured the share of news devoted to a given risk, while Burst Detection identified periods in which that attention increased unusually strongly relative to the theme's recent history.

The distinction between these measures was central to the empirical analysis. An increase in Coverage does not necessarily imply that the reported information is new. A risk may receive sustained attention because newspapers repeatedly discuss developments that are already familiar. Conversely, Novelty may increase when new information appears even if the overall volume of reporting remains limited. Semantic Drift adds another dimension by identifying changes in the content associated with a risk category, while Burst Detection distinguishes exceptional increases in attention from ordinary fluctuations in Coverage.

The analysis of the five risk categories showed that these distinctions are empirically relevant. The indicators did not follow a common pattern across themes, and their movements often differed even within the same category.

Climate Risk displayed relatively persistent news attention, with several periods of increasing Coverage rather than a single dominant episode. Its Novelty was particularly high during 2018, indicating that changes in the informational content of climate-related reporting did not simply coincide with its periods of greatest overall attention.

Supply Chain Risk became substantially more visible in the news after 2020, with elevated Coverage during 2021 and much of 2022. Its Novelty, however, followed a more gradual pattern and remained relatively low around 2021–2022. This contrast illustrates how a theme can become increasingly

prominent without a corresponding increase in the distinctiveness of incoming articles.

Pandemic Risk showed a different trajectory, characterized by a major change in its representation in the news beginning in 2020. The limited number of relevant articles before that period also demonstrated an important measurement issue: semantic indicators based on very few observations can be unstable and must be interpreted cautiously. Once Pandemic Risk became widely represented, the indicators provided a more informative description of how reporting evolved.

AI-Related Risk remained comparatively marginal in the news for much of the sample before experiencing a pronounced increase in Coverage during 2023. This increase was also reflected in the detection of unusual attention episodes. However, average Novelty did not show an equally clear rise during the same year. The result indicates that the rapid expansion of reporting about a risk does not necessarily correspond to an equivalent increase in semantic novelty.

These differences support the use of a multidimensional measurement framework. The emergence of a risk in the news cannot be described adequately by a single indicator because attention, informational novelty, and semantic change do not necessarily evolve together.

Cyber Risk was examined in greater detail, both because of its relevance to the financial application and because its indicators showed several distinct episodes over the sample period. Its Coverage generally remained around 1–2% of monthly news, although this relatively stable baseline was interrupted by periods of stronger attention.

The monthly Burst Detection procedure identified three main Cyber Risk episodes. The first extended from December 2018 to November 2019, representing a sustained period of unusually high attention. A shorter episode occurred in January 2020. The third took place in February and March 2022 and was the strongest in terms of intensity.

February 2022 was particularly important because different indicators pointed to substantial changes during the same period. Cyber Coverage reached approximately 3.24%, its highest monthly value in the sample, while Semantic Drift reached approximately 0.154. The coincidence of these movements indicates that unusually intense reporting was accompanied by a substantial change in the semantic content associated with Cyber Risk.

A different pattern emerged towards the end of 2023. Cyber Novelty increased markedly, with the mean, median, and upper part of its distribution all reaching high values. At the same time, Coverage was relatively low and the period was not classified as a Burst episode. This suggests that incoming Cyber-related information had become more distinct from recent reporting without producing a comparable increase in the theme's overall share of news.

The comparison between these two periods provides one of the clearest findings of the descriptive analysis. In early 2022, Cyber Risk was characterized by an exceptional increase in attention accompanied by strong semantic change. In late 2023, the most prominent signal was the increased novelty of the information being published. The indicators therefore identified different forms of change within the same risk category.

The final part of the thesis investigated whether the constructed Cyber Risk indicators were associated with subsequent equity returns. Following the empirical approach developed in the literature on cyber-risk news and financial markets, the analysis considered weekly changes in Coverage, Novelty, and Semantic Drift, together with six weekly lags of portfolio excess returns.

Two equally weighted portfolios were examined. The Diversified portfolio contained 195 firms from different industries. The Attacked + Security portfolio contained 76 firms, including 66 companies classified as having experienced cyberattacks and 10 cybersecurity firms. This design allowed the same

indicators to be evaluated for a broad group of companies and for firms more directly connected to cybersecurity.

The empirical results showed a clear difference between the financial information contained in the three indicators. Coverage was the only measure whose six lagged changes were jointly significant in both portfolios when the indicators were examined individually. Under HAC(6) inference, the corresponding p-values were 0.0076 for the Diversified portfolio and 0.0012 for the Attacked + Security portfolio. Novelty and Semantic Drift did not reach the same significance threshold.

The importance of Coverage was also confirmed by the combined specifications. When all three indicators were included, the six Coverage lags remained jointly significant conditional on Novelty and Semantic Drift, with HAC(6) p-values of 0.0220 and 0.0082 for the two portfolios. By contrast, the additional contribution of Novelty and Semantic Drift was not statistically significant once Coverage was included.

These results indicate that the three indicators do not contribute equally to the prediction of weekly excess returns. Novelty and Semantic Drift remain useful for describing the informational evolution of Cyber Risk, but the financial regressions do not provide comparable evidence that they contain incremental return-predictive information over the sample examined.

The estimated Coverage coefficients were positive across all six lags in both portfolios. This means that increases in Cyber Risk Coverage were associated with higher subsequent weekly portfolio excess returns, conditional on the included lagged variables. The positive relationship should not, however, be interpreted as evidence that greater cyber-risk attention causes stock prices to rise. The regressions establish a conditional predictive association, but they do not identify the economic mechanism responsible for it.

The contribution of Coverage was also assessed by comparing the explanatory power of an autoregressive return equation with that of a model augmented by six lags of changes in Coverage. Adding Coverage increased adjusted R^2 by 3.64 percentage points for the Diversified portfolio and 4.43 percentage points for the Attacked + Security portfolio. The corresponding partial R^2 values were 5.26% and 6.17%, indicating that Coverage explained an additional portion of the variation left unexplained by the restricted financial models.

These in-sample results were complemented by an out-of-sample forecasting exercise. Using an expanding estimation window, the thesis compared one-week-ahead forecasts from an AR(6) benchmark with forecasts from the same model augmented by lagged Coverage. The evaluation included 172 forecasts for each portfolio.

The model including Coverage achieved a lower root mean squared forecast error in both cases. The RMSE reduction was 1.19% for the Diversified portfolio and 1.61% for the Attacked + Security portfolio, corresponding to out-of-sample R^2 values of 2.36% and 3.20%. The one-sided Clark–West tests also indicated statistically significant improvements, with HAC(6) p-values of 0.0172 and 0.0064. The forecasting gains were relatively modest in magnitude, but they were observed in both portfolios and were supported by the adjusted forecast-comparison tests. This strengthens the evidence that Cyber Risk Coverage contains information not fully captured by the portfolios' own recent returns, within the forecasting periods and specifications considered.

The final empirical tests examined whether the relationship between the news indicators and returns changed during periods of unusually intense Cyber Risk attention. For this purpose, the Burst Detection procedure was adapted to weekly data.

The weekly procedure identified six Burst episodes over 2014–2023. Within the common financial

regression sample, 19 of the 366 observations were classified as contemporaneous Burst weeks. During these weeks, average Coverage was substantially higher than in ordinary weeks, as expected from the definition of the indicator. Average Semantic Drift was also higher, while average Novelty was slightly lower. These comparisons were descriptive and were not accompanied by formal tests of the differences.

Interaction regressions were then used to investigate whether the estimated relationships between lagged changes in the indicators and portfolio returns differed following Burst periods. The tests did not identify statistically significant differences between the two regimes for Coverage, Novelty, or Semantic Drift. Given the small number of Burst observations, this finding does not establish that the relationships are identical. It indicates only that the estimated specifications did not provide sufficient evidence of a difference.

A further sensitivity check excluded the 19 contemporaneous Burst weeks from the Coverage return regressions. Coverage remained jointly significant in both portfolios, with HAC(6) p-values of 0.0072 for Diversified and 0.0005 for Attacked + Security. The result suggests that the predictive association identified in the full sample was not driven exclusively by the observations classified as contemporaneous Burst weeks. Nevertheless, the retained regressions could still contain lagged information originating during earlier Burst periods.

Taken together, the empirical findings lead to two main conclusions.

First, financial news can be used to construct complementary measures of how emerging risks evolve. The analysis across five categories demonstrates that changes in attention, novelty, and semantic content are related but distinct phenomena. The Cyber Risk application illustrates this particularly clearly: the exceptional attention and semantic change observed in early 2022 differed from the increase in informational novelty observed at the end of 2023. A framework based only on article frequency would not distinguish these patterns.

Second, the financial relevance of these measures is not uniform. In the Cyber Risk application, Coverage provides the strongest and most consistent evidence of an association with subsequent weekly equity excess returns. Its contribution remains statistically significant when the other semantic indicators are included, improves in-sample model fit, and produces modest but statistically supported out-of-sample forecasting gains. Novelty and Semantic Drift contribute to the measurement of risk-related information, but the financial analysis does not establish an equivalent incremental predictive contribution for these measures.

Several limitations qualify these conclusions. The analysis is based on articles from five newspapers and therefore measures risks as represented in this particular information environment, rather than all information available to market participants. The classification and semantic indicators depend on the selected methodologies and their implementation choices. Their interpretation can also become less reliable when a risk category is represented by relatively few articles.

The financial results are specific to Cyber Risk, the two constructed portfolios, the weekly sample, and the regression and forecasting specifications used in the thesis. They do not establish a causal effect of news attention on returns, nor do they demonstrate that the same predictive relationship would hold for other risk categories, portfolios, or periods. The limited number of Burst observations also restricts the conclusions that can be drawn about differences between ordinary and exceptional attention regimes.

These limitations suggest possible extensions of the analysis. The measurement framework could be applied to additional news sources and risk categories, while the financial tests could be repeated for different portfolios and sample periods. Further work could also investigate whether the relationship

between news-based measures and market outcomes varies with the type of risk, the characteristics of the firms, or the broader financial environment.

The central contribution of this thesis is therefore the connection between the measurement of emerging risks in financial news and an empirical assessment of their market relevance. The four indicators provide different perspectives on the evolution of risk-related information, while the Cyber Risk application shows that one of these perspectives, the intensity of news coverage, contains measurable predictive information for weekly portfolio excess returns. The results support the usefulness of news-based risk measurement while also showing the importance of distinguishing descriptive information about risk emergence from statistically supported financial relationships.